Assignment 4: Decision Tree Model

Business Interpretation Based on a Business Plan

AdVantage Growth Studio — Marketing Campaign

Course	MBAI 5310G-001 — AI Programming
Institution	Ontario Tech University
Lecturer	Zahra Atf
Dataset	marketing_campaign_decision_tree_dataset.xlsx

Task 1: Understand the Business Problem

Business Overview

AdVantage Growth Studio is a data-driven campaign advertising and marketing agency that supports businesses in planning, running, and improving digital marketing campaigns. The agency works with customer data, campaign engagement data, and sales outcomes to help clients understand which customers are most likely to respond positively to a campaign.

Business Element	Description
Business type	Data-driven campaign advertising and marketing agency
Main service	Helping small and medium-sized businesses design, target, and evaluate marketing campaigns
Main customers	Retailers, online stores, local service providers, and small consumer brands
Main business goal	Improve campaign conversion rates while reducing wasted advertising budget
Machine learning goal	Predict whether a customer is likely to convert after receiving a marketing campaign

What problem is the business trying to solve?

The key business problem is campaign inefficiency. Many businesses spend money sending campaigns to customers who are unlikely to respond. At the same time, they may fail to identify customers who are ready to buy but need the right message, offer, or channel. This creates two problems: wasted marketing budget and missed revenue opportunities.

- Some customers receive offers but do not convert, which wastes advertising money.
- Some customers would convert if the business targeted them more effectively.

- Marketing teams need a simple, interpretable model that explains why a customer may or may not convert.
- Managers need business-friendly insights, not only technical model results.

Machine Learning Decision

Machine learning can help the business decide which customers should receive follow-up marketing attention — such as a discount reminder, a sales follow-up, a personalized email, or a retargeting advertisement. It can also support campaign budget allocation across channels such as Email, Social Media, Search Ads, Display Ads, and SMS.

Question	Answer
What is the target variable?	Converted
Possible target values?	Yes = customer converted after the campaign; No = customer did not convert
What decision can ML support?	Identify customers more likely to convert after a campaign
Who may use the model?	Marketing managers, campaign analysts, and client-facing strategists
What should be excluded?	Customer_ID and Campaign_Date — identifiers with no predictive value
Kind of model required?	Decision Tree classification model

Input Features

- Customer demographics: Age, Annual_Income, Region
- Customer relationship: Customer_Segment, Loyalty_Score, Customer_Satisfaction
- Campaign information: Campaign_Channel, Campaign_Type, Discount_Offered_Percent, Ad_Exposure_Count
- Engagement behavior: Email_Opened, Ad_Clicked, Website_Visits_Last_30_Days, Social_Media_Engagement_Score
- Purchase history: Previous_Purchases, Average_Order_Value, Days_Since_Last_Purchase, Prior_Campaign_Response

Why is this prediction useful?

Predicting conversion is useful because it turns campaign data into action. A marketing manager can use the prediction to decide where to spend money, which customers need follow-up, and which campaign features may be most effective.

- Better targeting: focus on customers with a higher probability of conversion.
- Budget control: reduce spending on customers who are unlikely to convert.
- Campaign improvement: learn which channels, offers, and engagement behaviors are linked to better results.
- Customer experience: avoid over-contacting customers who are unlikely to respond.

- Business communication: explain results to non-technical managers using feature importance and tree rules.

Task 2: Prepare the Data

Dataset Overview

The dataset was loaded from the provided Excel file. It originally contains 605 rows and 21 columns. The business plan notes that the dataset includes intentional duplicate rows and missing values for cleaning practice.

Metric	Result
Original rows x columns	605 x 21
Duplicate rows found	5 (removed)
Missing: Annual_Income	10
Missing: Customer_Segment	6
Missing: Average_Order_Value	6
Missing: Customer_Satisfaction	8
Rows after cleaning	600
Target distribution	309 No (52%), 291 Yes (48%) — well balanced

Data Cleaning

A copy of the original dataset was created. Customer_ID and Campaign_Date were removed as they are not predictive features. 5 duplicate rows were removed. Missing values in numerical columns were filled with the median value of each column. Missing values in categorical columns were filled with the most frequent value (mode). After cleaning: 600 rows, 19 columns, 0 missing values.

Feature Engineering

Categorical features were one-hot encoded using pd.get_dummies(). The target variable was label-encoded (No = 0, Yes = 1). After encoding, the feature matrix expanded from 18 to 35 features. The data was split 80/20 into training (480 records) and testing (120 records) using stratified sampling.

Task 3: Train the Decision Tree Model

Introduction to Decision Trees

A Decision Tree is a machine learning model that makes predictions by learning simple decision rules from the data. Each node in the tree asks a yes/no question about a feature, and each branch leads toward a final prediction. Decision Trees are useful for AdVantage Growth Studio because they are interpretable — marketing managers can read the rules and explain why a customer is predicted to convert or not.

How Decision Trees Make Predictions

Starting from the root node, the tree splits data based on the feature that best separates converters from non-converters. For example, the tree might first ask: Is the Loyalty_Score less than 64.5? Depending on the answer, it continues — perhaps checking Days_Since_Last_Purchase or Ad_Clicked — until reaching a leaf node that gives a final prediction of Yes (will convert) or No (will not convert). A tree that grows too deep memorizes training patterns and fails to generalize to new customers — this is overfitting.

Model Training and Evaluation

The Decision Tree was trained without depth constraints. The tree grew to a depth of 16 with 108 leaves.

Metric	Result
Training Accuracy	100.0%
Testing Accuracy	52.5%
Gap	47.5 percentage points
Precision (Yes class)	50.98%
Recall (Yes class)	44.83%
F1-Score (Yes class)	47.71%
True Negatives (TN)	37 — correctly predicted No
False Positives (FP)	25 — predicted Yes, actually No
False Negatives (FN)	32 — predicted No, actually Yes
True Positives (TP)	26 — correctly predicted Yes

Controlling Decision Tree Complexity

Metric	Result
Training Accuracy (max_depth=4)	72.9%
Testing Accuracy (max_depth=4)	58.3%
Gap (max_depth=4)	14.6 percentage points

Limiting tree depth reduced the gap from 47.5 to 14.6 percentage points and improved testing accuracy from 52.5% to 58.3%, confirming severe overfitting in the unconstrained model.

Top Feature Importances

Metric	Result
Loyalty_Score	19.4%
Days_Since_Last_Purchase	13.2%
Annual_Income	9.6%
Social_Media_Engagement_Score	7.9%
Average_Order_Value	7.7%
Previous_Purchases	7.5%
Website_Visits_Last_30_Days	5.9%
Age	4.3%
Discount_Offered_Percent	3.8%
Customer_Satisfaction	2.9%

Task 4: Business Interpretation of Model Results

How well did the model perform?

The Decision Tree achieved a testing accuracy of 52.5% on the 120-record test set. The precision for the Yes class is 50.98%, recall is 44.83%, and F1-score is 47.71%. This performance is barely above random guessing for a balanced dataset. With max_depth=4, testing accuracy improved to 58.3%. The model provides a starting point for targeting but requires significant improvement before being used in a real campaign.

Overfitting Analysis

The unconstrained model achieved 100% training accuracy but only 52.5% testing accuracy — a gap of 47.5 percentage points. The tree grew 16 levels deep with 108 leaves, memorizing every training record rather than learning generalizable patterns. This is severe overfitting. Applying max_depth=4 reduced the gap to 14.6 points, demonstrating that controlling complexity is essential.

Model Outcome and Business Meaning

Error Type	Business Meaning for AdVantage Growth Studio
False Positive (25)	The model predicts the customer will convert, but they do not. The business spends money on follow-up marketing for a customer who does not buy. This wastes advertising budget and staff time.
False Negative (32)	The model predicts the customer will not convert, but they actually do. The business misses a valuable customer who could have been targeted more effectively. This leads to lost revenue opportunities.
True Positive (26)	The model correctly identifies a customer who will convert. This supports effective and efficient campaign targeting.
True Negative (37)	The model correctly identifies a customer who will not convert. This avoids unnecessary outreach and protects the campaign budget.

Which evaluation metric matters most?

As stated in the business plan, F1-score is the recommended primary metric for this business problem because AdVantage Growth Studio wants to balance two risks equally:

- Precision matters when the business wants to avoid wasting campaign budget on unlikely converters.
- Recall matters when the business wants to find as many potential converters as possible and avoid missing them.
- F1-score balances both precision and recall and is recommended when both wasted budget and missed revenue matter equally.

The current F1-score of 47.71% shows the model is performing poorly on both fronts — it is missing too many true converters (low recall) and flagging too many non-converters (low precision).

Expected Important Features — Business Interpretation

Feature	Business Meaning
Loyalty_Score (19.4%)	Loyal customers are more likely to respond to campaigns. High loyalty signals a strong relationship with the brand.
Days_Since_Last_Purchase (13.2%)	Customers who purchased recently are still engaged. Customers inactive for a long time may need a stronger incentive.
Annual_Income (9.6%)	Income level relates to purchasing power and product affordability.
Social_Media_Engagement_Score (7.9%)	Active social media engagement signals customer interest in the brand.
Previous_Purchases (7.5%)	Repeat buyers are more likely to convert again — strong signal of purchase behavior.
Website_Visits_Last_30_Days (5.9%)	Frequent recent website visits indicate purchase intent.
Discount_Offered_Percent (3.8%)	Discounts may increase conversion but can also reduce profit margins.
Prior_Campaign_Response	Past response to campaigns can signal future responsiveness.
Ad_Clicked	Clicking an ad is a strong sign of customer interest — even if not in the top 10, it has direct campaign relevance.

How the Business Can Use the Model Results

Model Insight	Possible Business Action
Loyalty_Score is highly important	Create special campaigns for loyal customers and retention campaigns for low-loyalty customers.
Days_Since_Last_Purchase is highly important	Design reactivation campaigns for customers who have been inactive for a long time.
Social_Media_Engagement_Score is important	Focus social media retargeting on high-engagement customers.
Discount_Offered_Percent is important	Test different discount levels to balance conversion rate and profitability.
Campaign_Channel is in the top features	Move more budget toward channels with better conversion patterns.
Website_Visits is important	Retarget customers who visited the website recently but did not purchase.
Previous_Purchases is important	Design exclusive repeat-buyer offers to maintain loyalty and drive further conversions.

Limitation and Bias

The dataset is synthetic, does not represent all customer groups equally, and may not capture every factor that influences real customer behavior. As the business plan notes:

- Income and region may introduce fairness concerns if used without careful interpretation.
- The model can identify patterns but does not prove that one feature causes conversion.
- Missing values and duplicate rows can affect the model if they are not handled carefully — this was addressed during data cleaning.
- A Decision Tree can overfit if it becomes too deep and memorizes training data — confirmed by the 47.5 percentage point gap.

Limitations of Tree-Based Models

- Overfitting: Without constraints, Decision Trees grow too deep and memorize training data, leading to poor performance on new customers.
- Instability: Small changes in training data can produce a completely different tree structure.
- Limited generalization: A tree trained on 600 synthetic records cannot capture the full complexity of real customer behavior.
- Correlation vs causation: The model identifies patterns in the data but does not prove that any feature directly causes conversion.
- No probability calibration: The model predicts a class label but does not always produce well-calibrated probability scores.

Responsible AI Reflection

As the business plan states, the model should support human decision-making. It should not automatically replace marketing managers, customer service judgment, or ethical review.

- Fairness: Income and region may create unequal targeting patterns. Lower-income customers could be systematically excluded from campaign outreach even if they are genuinely interested and would convert given the right offer.
- Transparency: The Decision Tree is interpretable — managers can follow the decision rules and explain predictions to clients. This supports trust and accountability.
- Human oversight: Managers should review whether model recommendations make business sense, check for bias, consider customer privacy, and avoid excessive targeting.
- Ethical responsibility: The final decision should always combine model results, business goals, and responsible human oversight.

Final Conclusion

This report documented the full process of building and evaluating a Decision Tree classification model to help AdVantage Growth Studio predict which customers are likely to convert after receiving a marketing campaign.

The unconstrained model achieved 100% training accuracy but only 52.5% testing accuracy, showing severe overfitting with a 47.5 percentage point gap. When depth was limited to max_depth=4, testing accuracy improved to 58.3% with a gap of 14.6 points. The top predictors — Loyalty_Score, Days_Since_Last_Purchase, and Annual_Income — confirm that customer loyalty, purchase recency, and financial profile are the strongest signals of conversion likelihood.

As recommended in the business plan, F1-score is the primary metric because both wasted budget (false positives) and missed revenue (false negatives) are equally costly for this business. With 25 false positives and 32 false negatives in the test set, the model currently underperforms on both fronts and requires hyperparameter tuning and a larger dataset before real-world use.

The model provides a useful decision-support tool for prioritizing campaign outreach, but marketing managers must review outputs, test different discount levels, allocate budget toward better-performing channels, and design reactivation campaigns for inactive customers — all guided by the feature importance findings and the business plan's recommended action table.